# IMPORTANT INSTRUCTIONS

**Please ensure below pointers are met while submitting the Idea PPT:**

- 1. Kindly keep the maximum slides limit up to six (6). ( Including the title slide)
- 2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
- 3. Keep your explanation precise and easy to understand
- 4. Idea should be unique and novel.
- 5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
- 6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

**Note - You can delete this slide (Important Pointers) when you upload the details of your idea portal.**

# TITLE PAGE

- **Problem Statement ID –** [Insert Your PS ID e.g. SIH1234 / HW_SEC_01]
- **Problem Statement Title-** Autonomous Behavioral Anomaly Detection & Threat SOC Engine for Enterprise IT & Industrial OT
- **Theme-** Cybersecurity / AI & ML / Smart Industrial Automation
- **PS Category-** Software
- **Student Name (Registered on portal)-** Yash Singh
- **Student ID-** [Insert Your Registered Portal Student ID]

# IDEA TITLE: BehaviorIQ

## Proposed Solution (Describe your Idea/Solution/Prototype)

- **Detailed explanation of the proposed solution:**
    - **BehaviorIQ Engine:** Autonomous AI-powered SOC engine replacing static SIEM alert rules with dynamic, per-entity ML baselines.
    - **Dual-Stage ML Core:** Unsupervised Isolation Forest (5,000 decision trees) for anomaly scoring [0-100] + LightGBM Multi-Class Classifier.
    - **Explainable AI (SHAP XAI):** SHAP feature attributions and natural-language narratives for every alert.
- **How it addresses the problem:**
    - **99% Noise Reduction:** Enforces strict Top 1% Alert Budget (>=75.0 Risk Score), eliminating analyst alert fatigue.
    - **Sub-Second Mitigation:** Autonomous playbooks (Cloudflare WAF IP block, OAuth2 token revocation) in <0.38 seconds.
    - **Edge Cases:** Bayesian Blending for Cold-Start (0.85% FPR) and EMA (alpha=0.05) for Concept Drift.
- **Innovation and uniqueness of the solution:**
    - **Geo-Velocity Tracking:** Calculates physical travel speed (>840 km/h) between tokens to stop impossible travel.
    - **Dark Web Intelligence:** Live cross-referencing against active dark web credential dumps.

# TECHNICAL APPROACH

- **Technologies to be used (e.g. programming languages, frameworks, hardware):**
  - **Languages & Backend:** Python 3.11, FastAPI REST, Uvicorn, Gunicorn.
  - **ML & XAI:** Scikit-Learn (Isolation Forest), LightGBM, SHAP, NumPy, Pandas.
  - **Streaming & Cache:** Apache Kafka Broker (1,420+ eps), Redis Feature Store Cache (<0.2ms latency).
  - **Frontend & Cloud:** Glassmorphism HTML5 Canvas, Tailwind CSS v3, Chart.js, Docker, K8s HPA, Vercel, Render.
- **Methodology and process for implementation:**
  - **1. Ingestion:** Access Logs Stream (1,420 eps) ➔ Feature Engineering (Geo-Velocity, Failed Bursts).
  - **2. Profiling:** Feature Matrix ➔ Isolation Forest Profiler (5,000 Trees) ➔ Anomaly Score [0-100].
  - **3. Alert Budget:** Score >= 75.0 ? Flag Threat : Update EMA Profile (alpha=0.05).
  - **4. XAI & SOAR:** LightGBM Classifier ➔ SHAP Attribution ➔ Cloudflare WAF Block (<0.38s).

# FEASIBILITY AND VIABILITY

- **Analysis of the feasibility of the idea:**
  - **Model Accuracy:** Binary Anomaly F1-Score = 0.961, Precision = 0.972, Recall = 0.951.
  - **Real-Time Latency:** Inference completes in <0.8ms per event, handling >12,000 events/sec throughput.
  - **Alert Budget Control:** Top 1% cutoff maintains False Positive Rate <0.32%.
- **Potential challenges and risks:**
  - **Cold-Start Challenge:** Newly onboarded users or IoT devices (N < 20 events) risking false positive storms.
  - **Concept Drift Risk:** Evolving employee shift patterns causing baseline drift over time.
- **Strategies for overcoming these challenges:**
  - **Hierarchical Bayesian Blending:** Blends population priors with empirical stats, reducing cold-start FPR to 0.85%.
  - **EMA Baseline Adaptation (alpha=0.05):** Updates non-anomalous entity centroids without retraining.

# ARTIFACTS

- **Relevant artifacts, such as :**
- **Copy of the Code Embedded:**
  - **GitHub Repository:** https://github.com/Yash-Singh607/Behavior-IQ
  - **Containers & K8s:** Dockerfile, docker-compose.yml, deploy/k8s/deployment.yaml.
- **Snaps of the solution proposal & Live Cloud Links:**
  - **Live Vercel Frontend UI:** https://behavior-iq-jade.vercel.app
  - **Live Render FastAPI Backend:** https://behavior-iq.onrender.com
  - **Interactive Swagger API Docs:** https://behavior-iq.onrender.com/docs
  - **Analyst Testing Credentials:** Email: admin@behavioriq.ai **|** Password: admin123
- **Dashboard snaps:**
  - Real-Time Incident Triage Workbench with SHAP feature attributions.
  - Stage Threat Injection Simulator (■ Launch Attack Scenario trigger).
  - Glassmorphism Authentication Gate Modal & Profile Dropdown.

# RESEARCH AND REFERENCES

- **Details / Links of the reference and research work:**
  - **Isolation Forest Anomaly Detection:** Liu, F. T., Ting, K. M., & Zhou, Z. H. (2008). Isolation Forest. IEEE ICDM.
  - **LightGBM Gradient Boosting Framework:** Ke, G., et al. (2017). LightGBM. NeurIPS.
  - **SHAP Explainable AI Framework:** Lundberg, S. M., & Lee, S. I. (2017). SHAP. NeurIPS.
  - **MITRE ATT&CK; Framework:** MITRE ATT&CK; Matrix for Enterprise & ICS.
  - **Compliance Standards:** ISO/IEC 27001:2022, SOC 2 Type II, and GDPR Regulations.